

# HASID

## Smart Educational Investment Platform

### Frontend

- **React 18** — UI Library
- **TypeScript** — Type Safety
- **Vite 6** — Build Tool
- **Tailwind CSS 4** — Styling
- **Framer Motion** — Animations
- **Recharts** — Data Viz

Technology Stack

### UI Components

- **shadcn/ui** — Component System
- **Radix UI** — Accessibility
- **Lucide React** — Icons
- **Sonner** — Notifications
- **React Hook Form** — Forms
- **React Router 7** — Navigation

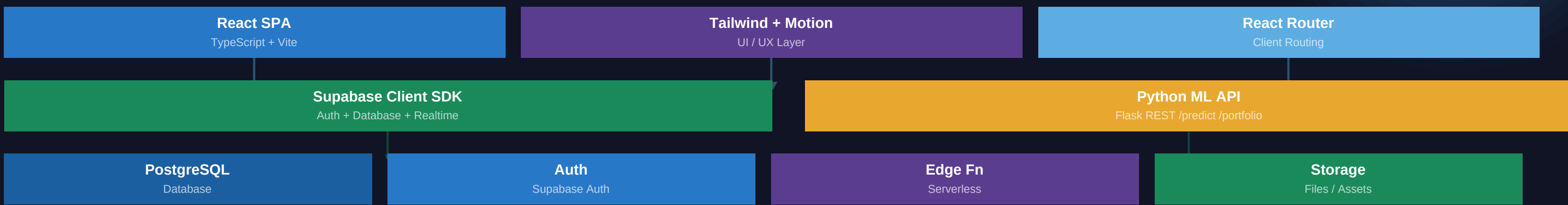
### Backend

- **Supabase** — BaaS Platform
- **PostgreSQL** — Database
- **Edge Functions** — Serverless
- **Row Level Security** — Auth
- **Real-time** — WebSocket
- **Storage** — File Management

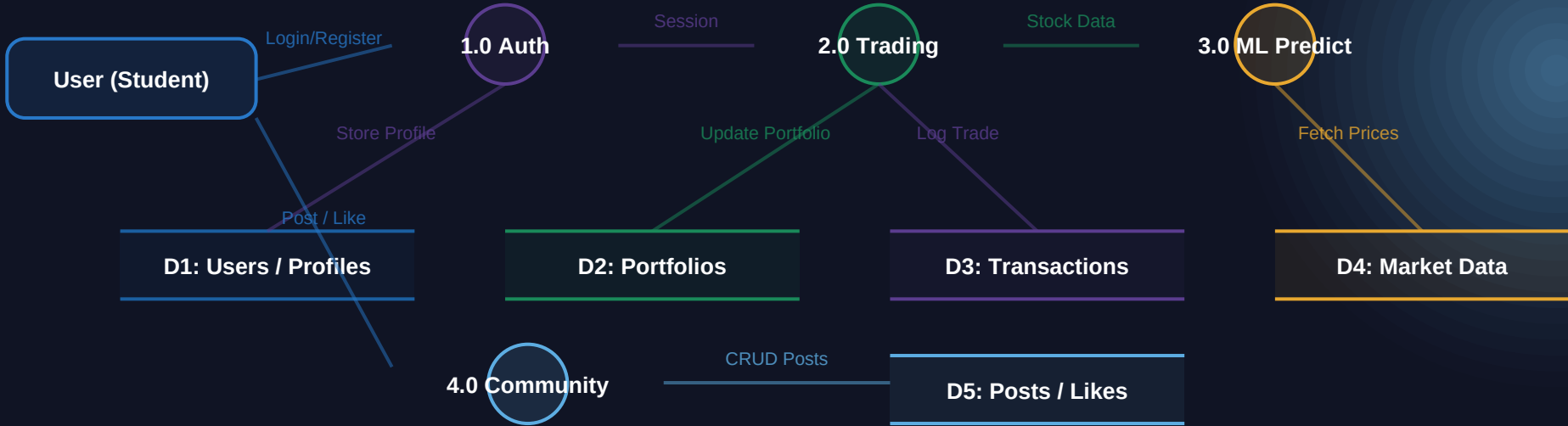
### ML / AI

- **Python Flask** — ML API Server
- **Scikit-learn** — ML Models
- **Linear Regression** — Prediction
- **Portfolio Optimizer** — MPT Algorithm
- **REST API** — /predict endpoint
- **NumPy / Pandas** — Data Processing

### System Architecture



### Data Flow Diagram



### Key Technical Features

#### Virtual Trading Engine

Real-time market data simulation, order matching engine, portfolio tracking, risk management tools.

#### Portfolio Optimization

Advanced algorithms for asset allocation, risk-adjusted returns, Monte Carlo Simulation (MPT).

#### Real-time Data

WebSocket connections for live price feeds, Supabase Realtime for user updates.

#### Dark Mode + Glassmorphism

Customizable UI themes, semi-transparent glass effects for modern aesthetics.

#### ML Stock Predictions

Machine learning models for price trend analysis, Python Flask API integration.

#### Gamification System

Points-based rewards for trading activity, educational content completion, leaderboard rankings.

#### Bilingual RTL/LTR

Support for right-to-left languages, automatic layout switching based on user preference.

#### Auth + RLS Security

Secure authentication flows, Row Level Security (RLS) for data isolation.

#### profiles

- id (UUID PK)
- email
- name
- avatar\_color
- level
- points
- balance

Database Schema

#### portfolios

- id
- user\_id (FK)
- stock\_symbol
- shares
- buy\_price
- created\_at

#### transactions

- id
- user\_id (FK)
- type
- symbol
- amount
- price
- timestamp

#### community\_posts

- id
- user\_id (FK)
- content
- likes
- comments
- created\_at

#### user\_achievements

- id
- user\_id (FK)
- badge\_id
- unlocked\_at
- progress

#### user\_settings

- id
- user\_id (FK)
- dark\_mode
- language
- notifications